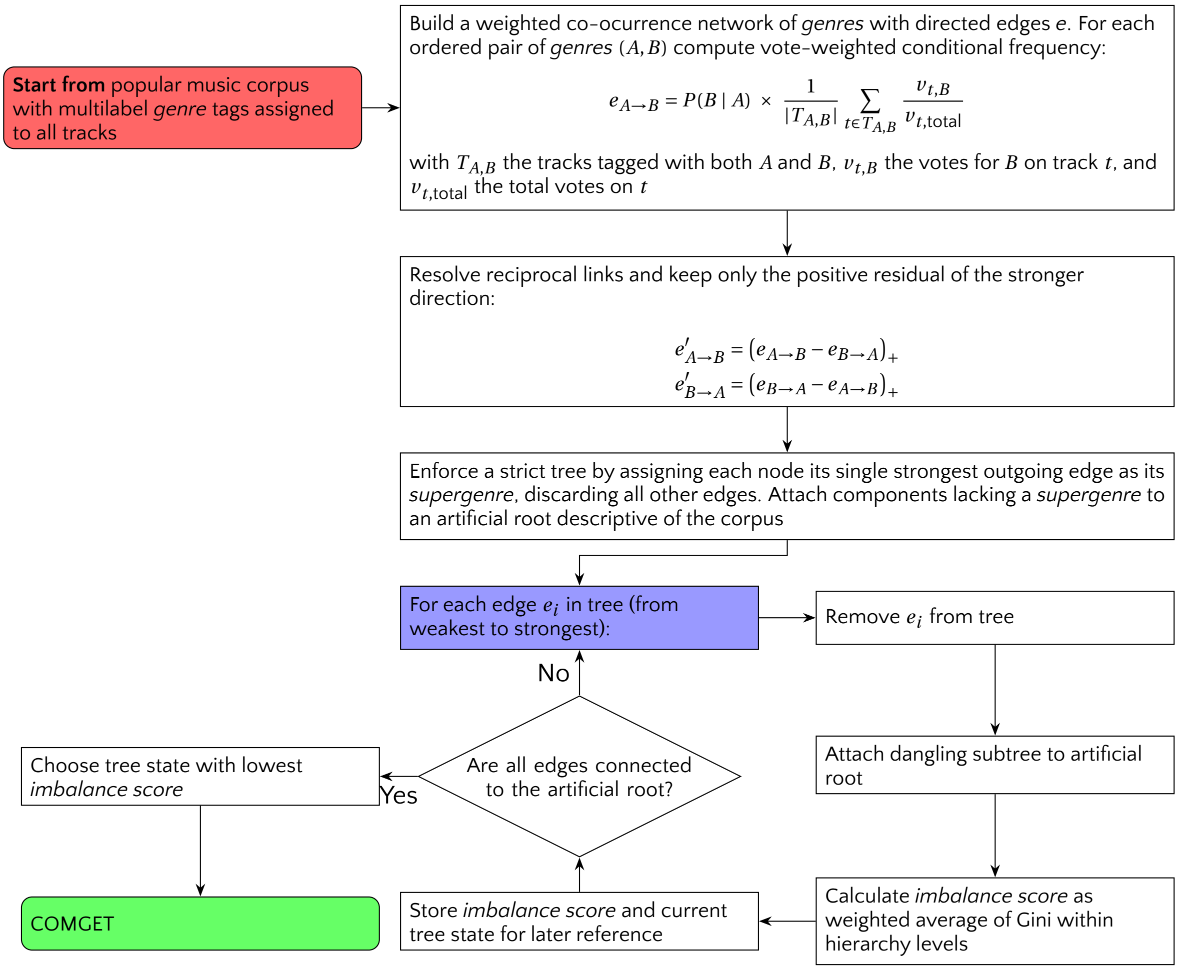
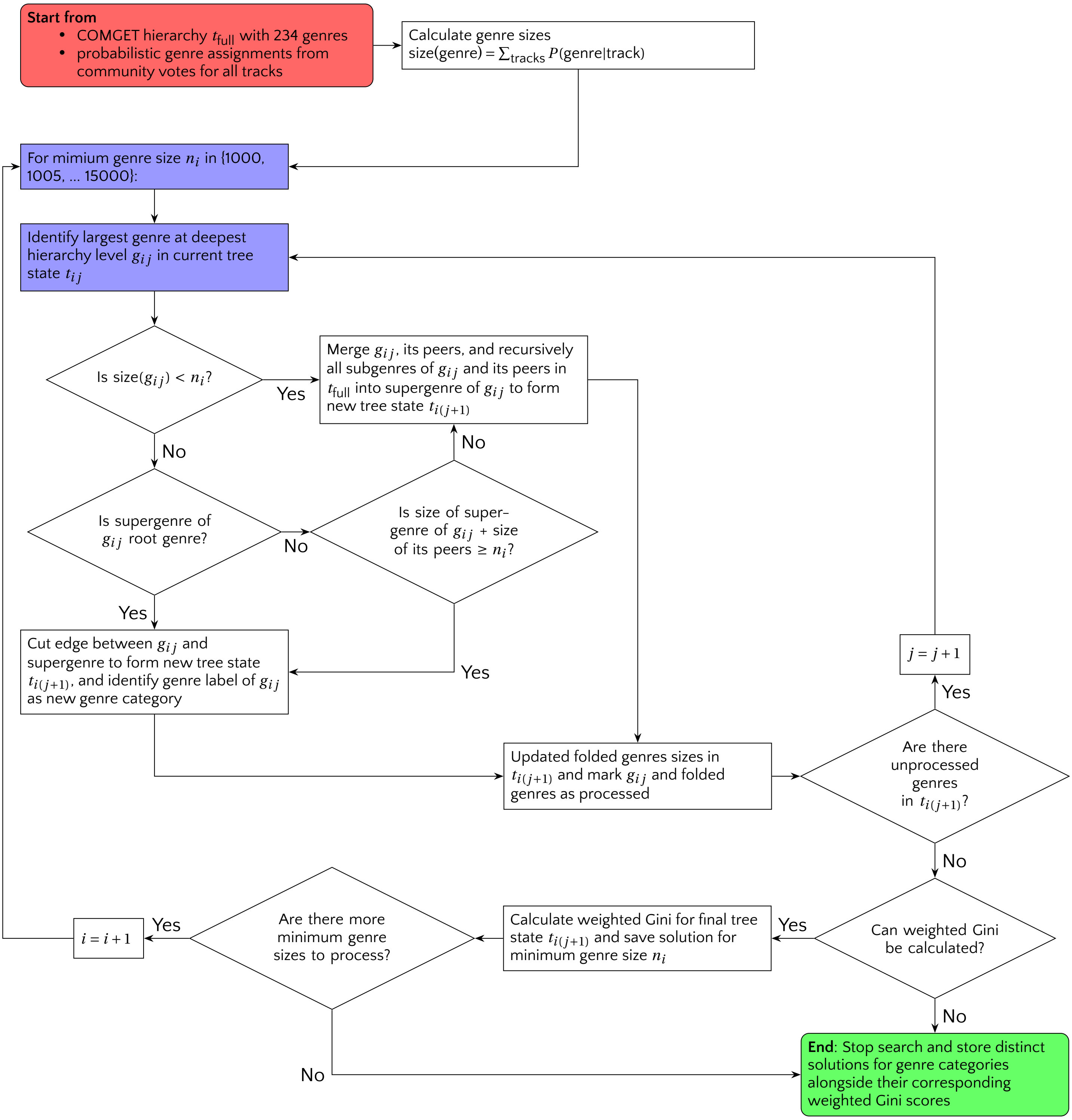
Appendix

A. RQ1 Detailed COMGET Algorithm For Constructing Balanced Hierarchical Trees

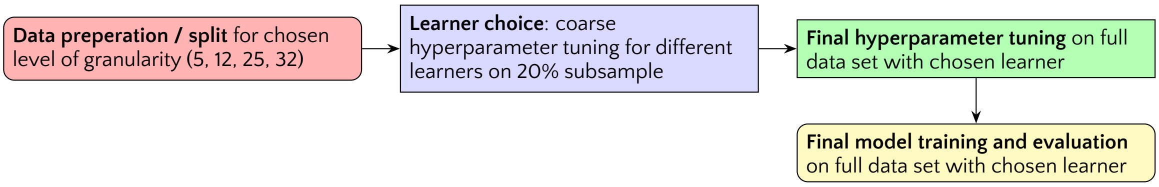


B. RQ2 Detailed FOLDCAT Flowchart for Folding COMGET to Identify Genre Categories

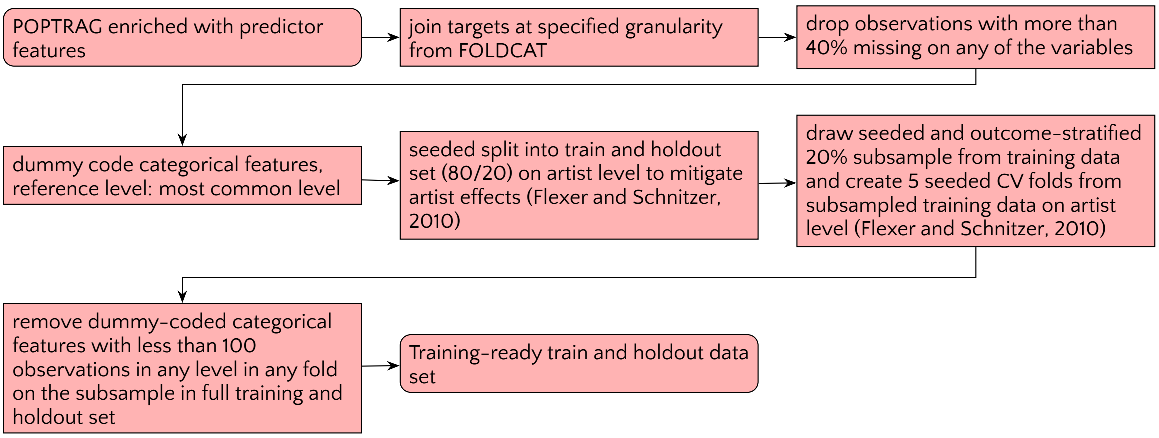


C. RQ3 Machine Learning Workflow

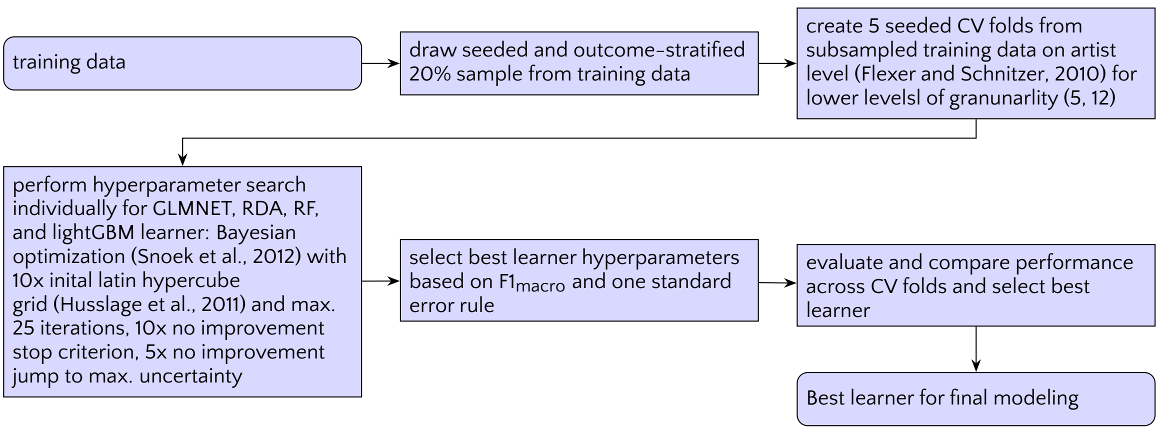
Overall Workflow



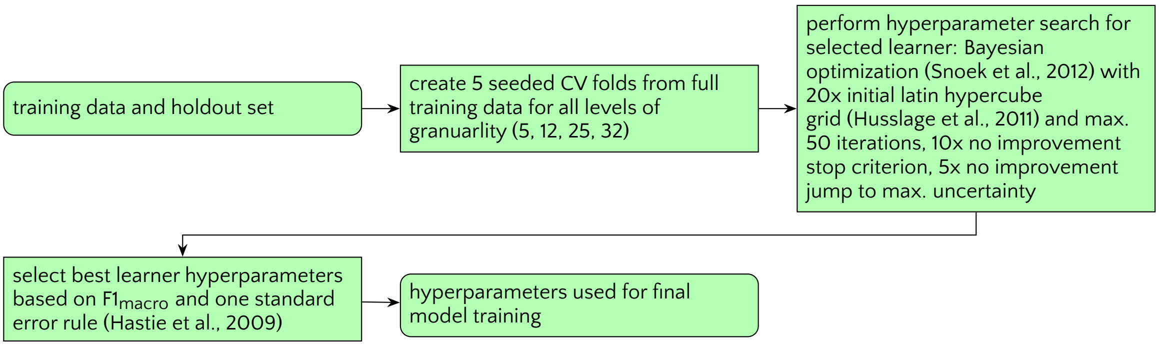
C.1 Data Preperation / Split



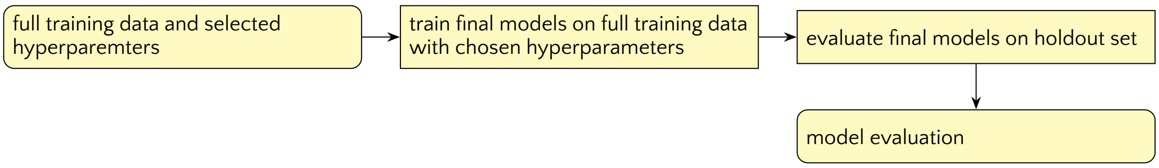
C.2 Learner Choice



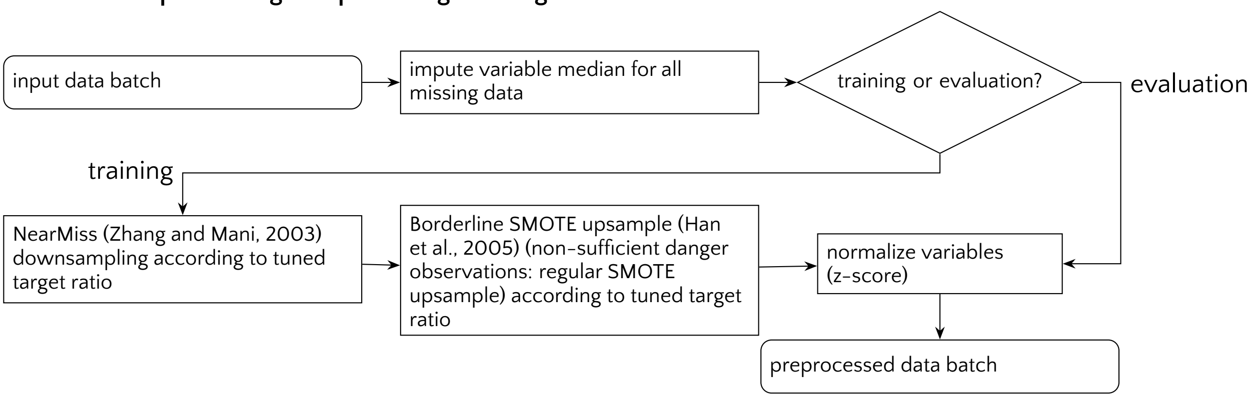
C.3 Final Hyperparameter Tuning



C.4 Final Model Training and Evaluation



C.5 Data Preprocessing Recipe During Training and Evalutation



Note on Adaptive Class Balancing During Data Prepreprocessing
Adaptive class balancing employed a single tunable *target-ratio* parameter governing rebalancing degree. The algorithm computed a reference count from outcome class frequencies (median for ≤ 10 classes; 60th percentile for 11–20 classes; 65th percentile for > 20 classes), then downsampled classes exceeding reference × *target ratio* and upsampled classes below reference ÷ *target ratio*. This approach preserved more data than fixed-ratio sampling while remaining tunable: *target ratio* = 1 enforced perfect balance; higher values retained original imbalance. The reference-count heuristic scaled to problem complexity, preventing excessive downsampling at higher resolutions.

D. RQ3 Comparison COMGET-G25 Against *Deezer*

COMGET-G25

$\text{Gini}_{\text{COMGET}} = 0.603$
$\text{Gini}_{\text{Deezer}} = 0.822$

Deezer First Genre	rock (21%)	metal (8%)	death metal (1%)	heavy metal (3%)	power metal (1%)	alternative rock (4%)	pop rock (2%)	hard rock (3%)	indie rock (3%)	punk (2%)	progressive rock (2%)	folk (1%)	folk rock (1%)	classic rock (1%)	blues rock (1%)	singer-songwriter (1%)	pop (18%)	electronic (10%)	jazz (3%)	schlager (3%)	soul (1%)	classical (1%)	synth-pop (1%)	r&b (1%)	hip hop (11%)
Rock (28%)	36	16	2	6	3	5	2	8	1	3	4	0	2	1	3	0	4	2	0	0	0	0	0	0	0
Heavy Metal (5%)	6	57	9	14	8	0	0	1	0	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
Alternative (12%)	33	2	0	0	0	12	1	0	15	7	1	1	1	0	0	1	9	10	0	1	0	0	1	0	1
Country (1%)	89	0	0	0	0	0	0	1	0	0	0	1	1	0	0	1	5	0	0	0	0	0	0	0	3
Pop (29%)	18	1	0	1	0	2	4	1	1	1	2	1	1	1	0	1	42	8	3	2	2	1	2	1	3
Electro (4%)	3	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	6	82	1	0	0	0	3	0	2
Dance (2%)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	85	0	0	1	0	0	0	3
R&B (2%)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	33	3	3	0	26	0	0	24	10
Jazz (2%)	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	3	89	1	1	1	0	1	0
Schlager & Volksmusik (3%)	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	3	14	0	1	80	0	0	0	0	0
Rap/Hip Hop (10%)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	0	0	1	0	0	1	92
Reggae (1%)	4	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	86	3	0	0	2	0	0	0	3

75%
50%
25%
0%

Figure D.1: Contingency tables comparing COMGET-G25 to release genres from the *Deezer* API; 27 genres were found in the data set; values show $P(\text{COMGET} \mid \text{External})$. Discarded 15 genres for visualization with less than 0.5% relative freuqency: *African Music, Asian Music, Bolero, Brasilian Music, Classical, Disco, Film / Video Games, Folk, German Music, Kids, Latin Music, Ranchera, Singer & Songwriter, Soul & Funk, Turkish Folk Music*.

E. RQ3 Features

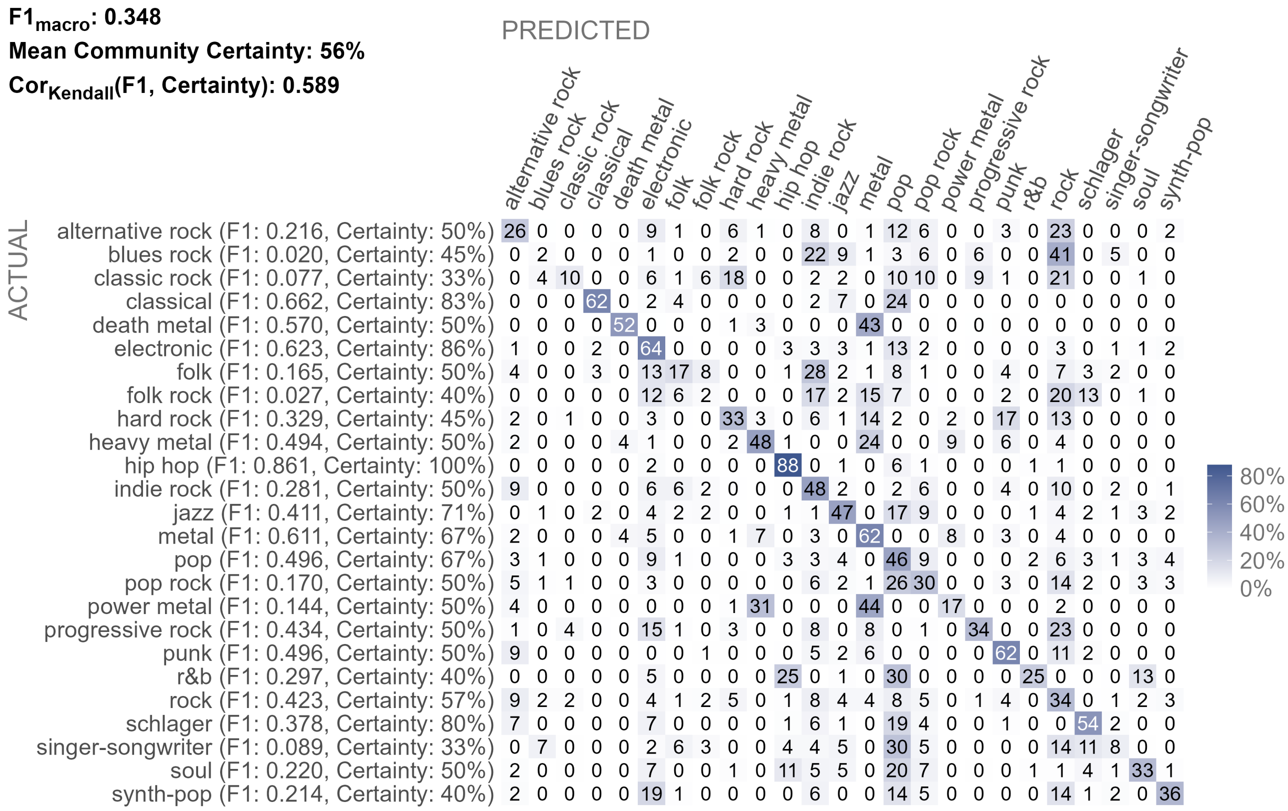
Construct	Subdimensions	Data Type	Indicators	Variable Content	Source
genre of a popular music track	release milieu (via metadata)	metadata release	artist	gender persona country of activity country of origin birth year is dead?	*MusicBrainz*
				27 Spotify distribution genres	*Spotify*
			release	release year indie label? D-A-CH label? release length	*MusicBrainz* *Spotify*
		metadata lyrics	code system	language	*Genius / MusicBrainz*
				track explicit?	*Spotify*
			information content	distinct words ratio repeated lines ratio	*Genius*
	musical style (via content analysis)	lexial content analysis of lyrics	emotion	positive negative anger disgust fear joy sadness suprise anticipation trust	*Genius* + *syuzhet*
			topics	sentiment	*Genius* + *sentimentr*
		machine-learning-based audio signal analysis	sound of production	loudness brightness acousticness liveness typicality genre sounds	*Spotify* (*Echo-Nest*) *Acousticbrainz* (*Essentia*)
			instrumnetation	instrumentalness tonality voice and female voice sound electronic sound	
			composition	tempo chords structure duration time signature key rhythm danceability emotional expression mood	
	popularity (via sucess metrics)	streams, followers, and release numbers on commercial music streaming platforms	track	popularity	*Spotify*
				rank	*Deezer*
			artist	streams	*Spotify*
				followers number of releases	*Deezer*
			label	median track populariy median release popularity median track popularity	*Spotify*

Table E.1: Features used for training the automated classifier and their theoretical dimension, adapted from anonymous autors (XXXX).

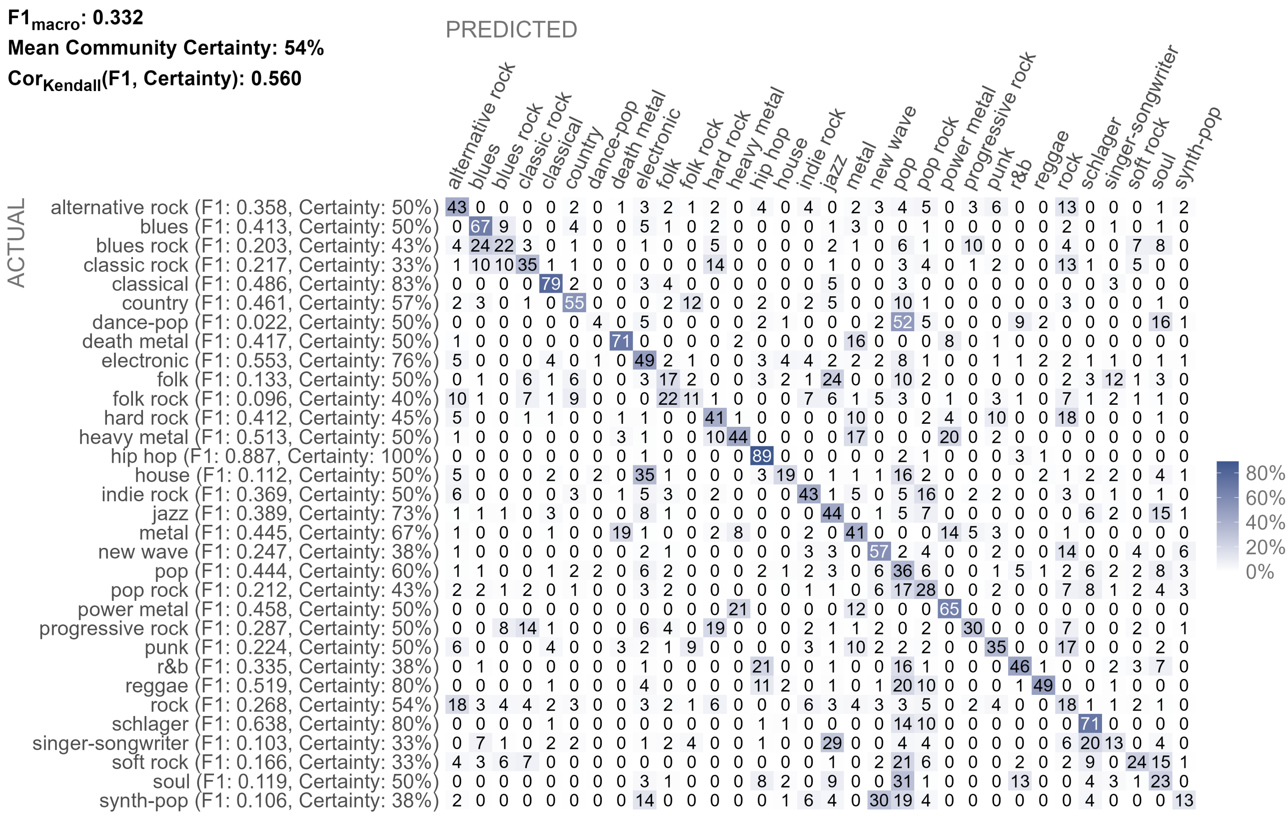
F. RQ3 Classification Hyperparmeters and Outcomes for Higher Granularities

# genres	boosting rounds [50, 2000]	tree depth [3, 15]	feature fraction [1, # of predictors]	leaf size [1, 100]	learning rate [0.001, 0.1]	target ratio [1, max/min]
5	93	12	68	46	0.072	4.815
12	1145	10	5	28	0.046	2.116
25	1512	9	85	91	0.005	2.641
32	176	9	55	15	0.0182	4.589

Table F.1: Hyperparameter settings for final LightGBM model training at different granularity levels.

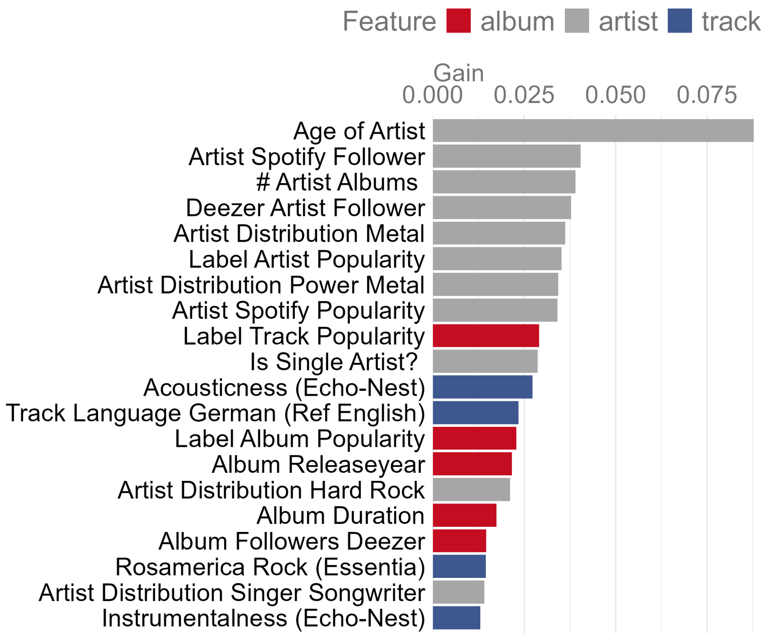


(a) COMGET-G25

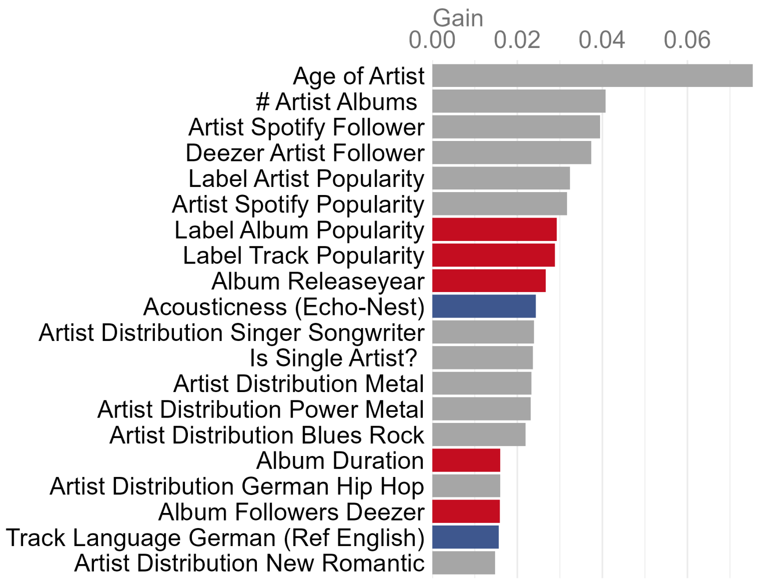


(b) COMGET-G32

Figure F.2: Confusion matrices for the final LightGBM genre classification models for COMGET-G25 and COMGET-G32 on the holdout set. Values show $P(\text{Predicted}|\text{Actual})$; *certainty* indicates the mean assignment probability for each *genre category* across community votes.



(a) COMGET-G25



(b) COMGET-G32

Figure F.3: Top 20 most important features for algorithmic genre classification of POPTRAG corpus at high (25) and very high (32) granularity of COMGET-G. Values show relative feature importance extracted from final LightGBM models.

G. Github Repository

Please find further details and the complete code for reproduction hosted on *Github*:
Link is left out for the purpose of anonymization of authors. Please find an interactive COMGET explorer .html file attached to this submission.
Data is available from the authors on request.

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